

Alan Mamulski

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TECHNICAL SKILLS

- **Languages & Frameworks:** Elixir, Phoenix, TypeScript, JavaScript, React, Node.js, GraphQL (Absinthe, Subgraph Federation), SQL
- **AI & Data Engineering:** Agentic AI (N8N), LLMs, Braintrust Evals, Prompt Engineering, Kafka, Flink, Debezium
- **Testing & Tools:** ExUnit, Jest, React Testing Library (RTL), Playwright, Webpack, Module Federation, Tailwind CSS, Git
- **Architecture:** Microservices, Micro Front-Ends, Data Pipelines, API Design, Telemetry & Observability

FOUNDING PROJECTS

Homebaked (homebakedapp.com)

Co-Founder & Lead Engineer | *September 2025 – Present*

- **Engineering Ownership:** Co-founded Homebaked alongside product partners, leading full-stack technical architecture, code quality, and production readiness for a live multi-tenant bakery marketplace (Web & native iOS App Store release).
- **Unified API & Full-Stack Architecture:** Scaled the platform into a type-safe Next.js 15 serverless codebase (App Router, Server Actions) with a unified backend API serving both the web platform and native Expo React Native mobile client.
- **Event-Driven Messaging & Payments:** Built an event-driven notification engine (APNs push, transactional emails, Twilio SMS) and marketplace checkout/payout flows using Stripe Connect for split payments and seller cashouts.
- **DevOps & Telemetry:** Integrated Mapbox GL for proximity discovery, and established CI/CD pipelines via Github Actions, Vercel and EAS with Sentry monitoring.

PROFESSIONAL EXPERIENCE

Bill.com | Draper, Utah

Software Engineer II | *September 2025 – June 2026*

- **AI Agent Optimization:** Enhanced the transaction auto-categorization AI agent within production Kafka/N8N pipelines by integrating internal OCR services and conducting Braintrust eval experiments to optimize receipt text extraction prompts.
- **Enterprise Scaling & Observability:** Engineered scaling improvements and robust telemetry for auto-categorization workflows, ensuring end-to-end visibility and reliability across high-throughput production pipelines.
- **Feature Architecture & Delivery:** Designed system architecture and led development for a receipt generation system using payment processor transaction data (e.g., Marqeta), enabling admins to streamline reconciliation workflows.

- **API Design & Subgraph Federation:** Drove GraphQL API design for new services using subgraph federation and enterprise authorization standards to ensure alignment **with backend guild standards**.
- **AI Security:** Collaborated cross-functionally with security engineering teams to evaluate and mitigate prompt injection risks and security vulnerabilities in LLM-powered workflows.
- **Performance Engineering:** Partnered with software architects to refactor high-traffic legacy modules approaching scale limits, improving system reliability and performance.

Anglepoint | Lindon, Utah

Software Engineer L3 | October 2024 – September 2025

- **Full-Stack Development:** Built and maintained Elixir backend services and React micro front-ends for software asset management tools, improving data accuracy and operational efficiency.
- **Roadmap & Architecture Planning:** Managed competing stakeholder priorities during technical planning, designing modular system architectures that balanced engineering standards with shifting product requirements.

Divvy / Bill.com | Draper, Utah

Software Engineer II | August 2023 – October 2024

- **Data Pipeline Engineering:** Implemented high-volume Kafka and Flink pipelines to sanitize and enrich large-scale real-time data streams from Debezium connectors, guaranteeing system fault tolerance and data consistency across microservices.
- **Full-Stack Feature Delivery:** Spearheaded feature projects across the Elixir/Phoenix stack using GraphQL/Absinthe, collaborating closely with product managers and UX designers to deliver robust backend and frontend capabilities.
- **Third-Party Integrations:** Built webhook-based third-party integration modules in Elixir, enabling automated and accurate real-time data synchronization with a 91% match rate.

Software Engineer I | June 2022 – August 2023

- **Micro Front-End Architecture:** Maintained a React/TypeScript micro front-end architecture using Webpack and module federation, decoupling teams from the monolithic deployment queue and boosting delivery velocity.
- **High-Impact Initiatives:** Led technical planning and frontend execution for a major analytics initiative providing account insights to the top 70% of revenue-generating users with zero critical (P0/P1/P2) bugs.
- **Design Systems:** Developed performant and accessible React components for an enterprise UI library, standardizing design patterns and reducing development timelines across engineering squads.

Software Engineer Intern / Associate | January 2022 – June 2022

- Completed an intensive software engineering internship and converted to a full-time role based on engineering contributions and execution.

EDUCATION

- **Full-Stack Web Development Bootcamp** | University of Utah (2021)